

Selection on prior healthcare use: bias, target shift, and a screen-and-respond policy

A simulation study answering the conditional-mechanism part of catalog problem ELG-02

Ahmad Sofi-Mahmudi

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Abstract

Background. Restricting an emulated trial to people with prior healthcare use conditions on a variable that treatment propensity and outcome risk may both cause. Catalog entry ELG-02 states that such criteria are used without checking whether they are colliders. We measured the bias this restriction produces under known graphs, how much of the restricted-versus-broad contrast is a change of target, and whether screening an external sample for a collider signal improves estimation.

Methods. Eighteen generated cells crossed three eligibility topologies (noncollider, concordant collider, discordant collider), three edge strengths and two retention probabilities, with 2000 replicates each. Augmented inverse probability weighted estimators of the restricted 60-month risk difference were compared with a policy that screened a 4000-person audit sample and chose an estimator from the result, in a view where the collider parents were measured and one where they were not.

Results. The minimal estimator's absolute bias reached 0.036, with coverage down to 0.728. Five of the twelve collider cells had a lower bound of at least 0.010, spanning both orientations, both retention levels and two edge strengths: materially consequential by the registered rule, though the two-strength condition rests on one cell. With the parents measured the policy reduced bias in 7 of 12 collider cells, but adjusting for every measured variable without screening did as well, and pooling the audit sample into the analysis did better in every cell. With the parents unmeasured the screen never warned and the policy did nothing.

Conclusions. Selection on healthcare use can bias an emulation materially when the criterion is a collider. A screen adds nothing that adjusting for the measured variables does not already give, and it cannot see a collider whose parents are not recorded.

1 Introduction

An emulated trial restricted to people with prior healthcare use, such as a recent visit or an earlier prescription, conditions on a variable that treatment propensity and outcome risk may both cause. When they do, the restriction opens a path between them that the analysis model does not block (1). Catalog entry ELG-02 states that eligibility criteria built from healthcare use are applied without checking whether they are colliders. Restriction also changes the population the effect refers to, so a difference between restricted and unrestricted estimates mixes selection bias with a legitimate change of target (2). Reporting guidance asks that eligibility criteria and their emulation be stated (3); it does not say how to assess them.

This study answers the part of ELG-02 that simulation can answer. Given a known graph, how much bias does restriction on healthcare use produce, how much of the restricted-versus-broad contrast is target shift, and does a prespecified policy of screening an external sample for collider signals and responding to the result estimate the restricted effect better than analyses that do not screen? It does not determine whether any real eligibility criterion is a collider.

2 Methods

The study follows the ADEMP structure (4). The protocol, decision rule and code were registered before any replicate was drawn.

2.1 Data-generating mechanism

Each replicate draws an audit sample and an analysis sample of 4000 people each from one superpopulation. Baseline variables are $B \sim \text{Bernoulli}(0.5)$ and X, P, D and eight noise variables N_1 to N_8 , each standard normal, all mutually independent. P is a prescribing tendency that affects treatment only; D is preclinical severity that affects outcome only. Healthcare use H , treatment A and the 60-month outcome Y are probit:

- $H: \alpha_H + 0.30B + 0.20X + \lambda_P P + \lambda_D D$, with α_H solved for the retention probability;
- $A: -0.05 + 0.35B - 0.25X + 0.55P$;
- Y under treatment $a: -1.50 + 0.45B + 0.35X + 0.65D + a(-0.35 + 0.25B)$.

Eighteen cells cross three topologies, three edge strengths s (0.10, 0.45, 1.10) and two retention probabilities $\Pr(H = 1)$ (0.30, 0.70). The noncollider topology sets $\lambda_P = 0$ and $\lambda_D = s$, so H depends on outcome risk alone and restriction changes the population without opening a path. The concordant and discordant collider topologies set $\lambda_P = s$ with $\lambda_D = s$ or $-s$, so H depends on both parents and restriction to $H = 1$ induces a negative or positive association between P and D . There are twelve collider cells and six noncollider cells.

2.2 Estimands and truth

The primary estimand is the restricted 60-month risk difference $RD_H = E(Y^1 - Y^0 \mid H = 1)$. The broad estimand RD_{all} is the same contrast without restriction, and the target shift is $RD_H - RD_{\text{all}}$, a difference between two valid estimands rather than a bias. Both were computed by product Gauss-Hermite quadrature over X, P and D with exact summation over B , accepted when 30-, 40- and 50-node results agreed within 10^{-7} . The target shift was decomposed across B, X and D by an order-averaged chain-rule decomposition; the D component is an oracle diagnostic because D is unmeasured in the reduced view.

2.3 Methods compared

All causal estimators are augmented inverse probability weighted (AIPW) with probit nuisance models and a stacked M-estimation sandwich variance. Intervals are untruncated 95% Wald intervals.

- Minimal: restrict to $H = 1$; treatment model $A \sim B + X$; outcome model $Y \sim A + B + X + A:B$.
- Role-based: adds P to the treatment model and D to the outcome model, chosen from known covariate roles without screening. Unavailable in the reduced view.
- All-measured: every measured baseline variable in both models.
- Oracle: B, X, P and D in both models, including in the reduced view.
- Broad: criterion removed; estimates RD_{all} over all 4000 people and is compared only with RD_{all} .
- Pooled: the all-measured specification on audit and analysis samples together (8000 people), to price the audit sample.
- Policy: a screen on the audit sample, locked before the analysis sample is examined. Probit models for H, A and Y on B, X and every other measured variable as unlabeled candidates, Holm correction at familywise $\alpha = 0.01$ across candidates and models; the result is “collider candidate” if one candidate is associated with H and A and a different one with H and Y . On a collider-candidate result the policy reports the all-measured estimate, on “no signal” the minimal estimate, and on an unresolved audit the all-measured estimate with a flag.

Two analyst-information views are applied to every cell. In the full view P and D are measured. In the reduced view they are not, and the screen sees only the noise variables, so a hidden collider can pass unnoticed.

2.4 Decision rule

Collider bias is materially consequential if the lower 95% Monte Carlo bound for absolute bias of the minimal estimator is at least 0.010 in at least 4 of the 12 collider cells, the qualifying cells spanning both collider orientations, both retention levels and at least two edge strengths; it is negligible if the upper bound is below 0.005 in all 12. The policy is conditionally beneficial in a view if the lower 95% bound for the paired reduction in absolute bias relative to the minimal estimator, $|\text{Bias}_{\text{minimal}}| - |\text{Bias}_{\text{policy}}|$, is at least 0.005 in at least 4 spanning collider cells with no material harm in noncollider cells, and not beneficial if the upper bound is below 0.005 in all 12. The run is technically uninformative if a required estimator fails in more than 10% of replicates in any cell, fewer than 1900 paired outputs remain, the audit screen fails in more than 10% of replicates in any cell-view, quadrature fails its check, the broad estimator fails to recover RD_{all} , or more than one noncollider cell has an upper absolute-bias bound for the minimal estimator of at least 0.005. Monte Carlo bounds come from 2000 bootstrap resamples of replicates. Each cell has 2000 replicates, giving a Monte Carlo standard error of 0.0049 for coverage near 0.95.

3 Results

All 36,000 replicate pairs produced a valid estimate from every available method, no audit was unresolved, quadrature met its tolerance in every cell, and the broad estimator recovered RD_{all} . The run is technically informative.

3.1 The screen

With the parents measured, the screen warned in 0.754 to 0.777 of replicates at the near-null edge strength and in every replicate at strengths 0.45 and 1.10. In noncollider cells it warned in at most 0.0005. With the parents unmeasured it never warned, so every collider replicate in the reduced view was a silent error: the screen reported no signal and the policy used the biased minimal estimator.

3.2 Collider bias

Restriction biased the minimal estimator in the direction of the induced association: negatively in concordant cells and positively in discordant ones (Table 1). Absolute bias reached 0.036 in the concordant cell at strength 1.10 and retention 0.30, where coverage fell to 0.728. At the near-null strength it was at most 0.0011. In noncollider cells, where restriction changes the population without opening a path, the upper bound for absolute bias never exceeded 0.0024.

Five collider cells had a lower bound of at least 0.010: all four at strength 1.10 and the concordant cell at 0.45 with retention 0.30. They span both orientations, both retention levels and two edge strengths, so the registered classification is materially consequential. The two-strength requirement is met by that one cell, whose lower bound is 0.0112; without it the classification would be mixed.

Table 1: Minimal restricted estimator in the collider cells, full view: absolute bias with its 95% Monte Carlo interval, coverage, and the policy’s paired reduction in absolute bias with its lower bound.

topology	strength	retention	bias	interval	coverage	reduction	lower
concordant	0.10	0.3	0.0008	0.0001 to 0.0018	0.954	0.0002	-0.0004
concordant	0.10	0.7	0.0011	0.0004 to 0.0017	0.949	0.0002	-0.0001
concordant	0.45	0.3	0.0123	0.0112 to 0.0133	0.925	0.0117	0.0096
concordant	0.45	0.7	0.0065	0.0058 to 0.0071	0.923	0.0063	0.0051
concordant	1.10	0.3	0.0358	0.0346 to 0.0370	0.728	0.0358	0.0334
concordant	1.10	0.7	0.0180	0.0173 to 0.0187	0.768	0.0177	0.0169
discordant	0.10	0.3	0.0008	0.0001 to 0.0017	0.958	0.0007	-0.0009
discordant	0.10	0.7	0.0006	0.0001 to 0.0011	0.945	0.0003	-0.0003
discordant	0.45	0.3	0.0081	0.0073 to 0.0088	0.923	0.0077	0.0062
discordant	0.45	0.7	0.0059	0.0054 to 0.0064	0.937	0.0059	0.0048

Table 1: Minimal restricted estimator in the collider cells, full view: absolute bias with its 95% Monte Carlo interval, coverage, and the policy’s paired reduction in absolute bias with its lower bound.

topology	strength	retention	bias	interval	coverage	reduction	lower
discordant	1.10	0.3	0.0173	0.0166 to 0.0180	0.788	0.0169	0.0162
discordant	1.10	0.7	0.0152	0.0147 to 0.0157	0.732	0.0152	0.0142

3.3 The policy

With the parents measured, the policy’s reduction in absolute bias had a lower bound of at least 0.005 in 7 of 12 collider cells, spanning both orientations, both retention levels and two strengths, with no harm in the noncollider cells: conditionally beneficial by the registered rule. The one cell at strength 0.45 or above that missed was the discordant cell with retention 0.70, at 0.0048. With the parents unmeasured the screen never warned, the policy reported the minimal estimate in every replicate, and its benefit was exactly zero: not beneficial.

The benefit is not the screen’s. In the full-view collider cells the policy and the all-measured estimator, which uses the same variables without screening, differed in mean squared error by at most $2.4e-05$, and the policy was never the worse of the two. Pooling the audit and analysis samples and adjusting for every measured variable gave a lower mean squared error than the policy in 36 of 36 cell-views, each by more than three Monte Carlo standard errors. The registered design calls the policy data-efficient only if it beats that comparator, and it does not.

3.4 Target shift

Restriction also changes the estimand. The target shift $RD_H - RD_{all}$ ran from -0.019 to 0.013, carried almost entirely by D: its contribution reached 0.019 against at most 0.0021 for B and 0.0025 for X. Noncollider cells shift as much as collider cells, because H selects on outcome risk in both. A difference between restricted and unrestricted estimates is therefore not evidence of collider bias.

4 Discussion

Restricting on a healthcare-use variable that is a collider biases a correctly specified minimal analysis by up to 0.036 on the risk-difference scale in this benchmark, enough to move coverage from 0.95 to 0.73. It is at most 0.001 when the edges into the criterion are weak, 0.006 to 0.012 when they are moderate and 0.015 to 0.036 when they are strong, and it takes the sign of the association the selection induces. Restriction that selects only on outcome risk moves the estimand by a similar amount without biasing it, so the movement between a restricted and a broad estimate cannot distinguish the two.

The screen detects the collider when both parents are recorded, with almost no false alarms, but that is exactly the situation in which it is unnecessary: adjusting for every measured baseline variable removes the bias without it, and spending the audit sample on the analysis instead is better still. When the parents are not recorded the screen has nothing to find. A measured-variable screen of this kind therefore does not address the part of ELG-02 that matters, a criterion whose other causes are unrecorded.

4.1 What this study does not establish

The study is a mechanism benchmark under correctly specified probit models and a four-node graph that is known to the simulation. It does not discover or certify collider status for an unknown eligibility criterion, identify an unnamed latent cause, or estimate how often real criteria are colliders. The screen’s operating characteristics do not transfer to indirect paths, association-mimicking noncolliders, nonlinear mechanisms, noisy proxies, partial measurement or audit samples of other sizes. The two views are extremes: every parent measured perfectly, or neither measured at all. The policy’s performance assumes access to an independent audit sample of 4000 people.

4.2 Review

The design was reviewed and revised before any code was written. A first complete run was analyzed and discarded: its audit screen returned “unresolved” in every replicate, because it read standard errors from a coefficient table that `glm.fit` does not produce, so every policy estimate was the all-measured estimate and the policy branch it reported meant nothing. The screen was fixed, and a check that counts screen failures toward the registered failure ceiling was added, before the rerun reported here. The results have not yet been reviewed.

Data and code availability

Code, protocol and result files: <https://github.com/choxos/TTE-open-problems/tree/main/studies/ELG-02-baseline-healthcare-use-selection>.

References

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